Four Eggs, N Floors: Is 1–3 Better Than 2–2?

Abstract

With four eggs and N floors, the worst-case number of drops of any reasonable strategy is of order $N^{1/4}$, so the question "1–3 or 2–2?" is a question about constants. In the 1–3 approach we drop one egg at *optimally chosen* (non-uniform) intervals and, once it breaks, solve the remaining 3-egg problem optimally. We prove that this approach is in fact optimal among *all* four-egg strategies, and needs $24^{1/4}N^{1/4} + O(1) \approx 2.213\,N^{1/4}$ drops. The best 2–2 approach (equal superfloors of F floors, two eggs on the superfloors, two eggs inside the critical superfloor) needs $2\sqrt{2}\,N^{1/4} + O(1) \approx 2.828\,N^{1/4}$ drops. Hence 1–3 **is better**: it is never worse than 2–2 for any N, and for large N it is better by a factor $(8/3)^{1/4} \approx 1.278$ (that is, 2–2 needs about 28% more drops).

1 The problem and the ground rules

A building has N floors. We have k identical eggs. There is a number n (the *critical floor*) such that an egg dropped from any floor below n survives undamaged (and can be reused), while an egg dropped from floor n or higher breaks. If the egg survives even from floor N, we say $n = N + 1$. Our job is to determine n, and we measure a strategy by its *worst-case* number of drops over all possible values of n (the interpretation "minimize the worst case" from the talk [1]).

Since $n \in \{1, 2, \ldots, N + 1\}$, there are $N + 1$ possibilities. A drop from floor x splits the possibilities in two: if the egg breaks then $n \le x$, and if it survives then $n > x$.

Notation. Let $D_k(m)$ be the smallest number of drops that always suffices to find n when we have k eggs and m floors we are allowed to test (so there are $m + 1$ possibilities for n). Thus $D_k(N)$ is the answer to the original problem with k eggs. We set $D_k(0) = 0$.

The two strategies.

- **1–3.** Drop one egg from an increasing sequence of floors $x_1 < x_2 < x_3 < \cdots$ that we are free to choose (the *intervals* $x_1,\ x_2 - x_1,\ x_3 - x_2, \ldots$ need not be equal), stopping at the first drop in which the egg breaks. If it breaks at x_j, then $n \in \{x_{j-1} + 1, \ldots, x_j\}$ and we have 3 eggs left, so we solve this block with the best 3-egg strategy. The design question is how to choose the x_j optimally.

- **2–2.** Fix F and group the possibilities $n = 1, \ldots, N + 1$ into $M = \lceil (N + 1)/F \rceil$ consecutive blocks ("superfloors") $\{1, \ldots, F\}$, $\{F + 1, \ldots, 2F\}$, $\ldots$, each with at most F possibilities. Use the best 2-egg strategy on the $M - 1$ testable superfloor boundaries $F, 2F, \ldots$ to find the block containing n (at most two eggs are lost), then use the best 2-egg strategy on the $F - 1$ testable floors inside that block with the two remaining eggs.

In both strategies the sub-problems are solved *optimally*; in particular the 2-egg searches use the decreasing steps $x, x-1, x-2, \ldots$ (triangular numbers) discussed in the talk, not constant steps. So we first need to know D_2 and D_3 precisely.

2 How many floors can d drops handle with k eggs?

Rather than computing $D_k(m)$ directly, it is easier to flip the question around: given d drops and k eggs, what is the *largest* number $g(d,k)$ of floors that can be handled?

Lemma 1 (Capacity)**.** *For $d \geq 0$ and $k \geq 0$,*

$$g(d,k) = f(d,k) := \sum_{i=1}^{k} \binom{d}{i}.$$

That is, $D_k(m) \leq d$ if and only if $m \leq f(d,k)$.

Proof. Clearly $g(0,k) = 0$ (with no drops we can only handle $m = 0$ testable floors) and $g(d,0) = 0$ (with no eggs we cannot test anything). Now let $d, k \geq 1$ and suppose a strategy resolves m floors using at most d drops. Its first drop is from some floor $x \in \{1, \ldots, m\}$.

- If the egg breaks, $n \in \{1, \ldots, x\}$: we have $k-1$ eggs and $d-1$ drops left and $x-1$ testable floors, so $x - 1 \leq g(d-1, k-1)$.
- If it survives, $n \in \{x+1, \ldots, m+1\}$: we still have k eggs, $d-1$ drops, and $m-x$ testable floors, so $m - x \leq g(d-1,k)$.

Adding, $m \leq 1 + g(d-1,k-1) + g(d-1,k)$. Conversely, if m equals this bound, drop first from floor $x = 1 + g(d-1,k-1)$; both outcomes are then exactly manageable. Hence

$$g(d,k) = 1 + g(d-1,k-1) + g(d-1,k).$$

Now check that f satisfies the same recursion and boundary values. Clearly $f(0,k) = 0 = f(d,0)$. By Pascal's rule $\binom{d}{i} = \binom{d-1}{i-1} + \binom{d-1}{i}$, so

$$\begin{aligned} f(d,k) &= \sum_{i=1}^{k} \binom{d-1}{i-1} + \sum_{i=1}^{k} \binom{d-1}{i} \\ &= \Big[1 + \sum_{j=1}^{k-1} \binom{d-1}{j}\Big] + f(d-1,k) = 1 + f(d-1,k-1) + f(d-1,k). \end{aligned}$$

By induction on d, $g = f$. (Resolving fewer than $f(d,k)$ floors is of course no harder.) □

Examples.

- $k = 1$: $f(d,1) = d$. (One egg: go up one floor at a time.)
- $k = 2$: $f(d,2) = d + \binom{d}{2} = \dfrac{d(d+1)}{2}$, the triangular numbers. For example $N = 105 = \frac{14 \cdot 15}{2}$ needs exactly 14 drops, matching the example in the talk.

- $k = 3$: $f(d,3) = d + \binom{d}{2} + \binom{d}{3} = \dfrac{d^3 + 5d}{6}$.
- $k = 4$: $f(d,4) = \dfrac{d^4 - 2d^3 + 11d^2 + 14d}{24}$.

In each case the leading term is $d^k/k!$. The next lemma turns this into an explicit formula for the number of drops.

Lemma 2 (Size of $D_k(m)$)**.** *For all $k \geq 1$ and $m \geq 1$,*

$$\left| D_k(m) - (k!\, m)^{1/k} \right| \leq k.$$

In particular $D_2(m) = \sqrt{2m} + O(1)$, $D_3(m) = (6m)^{1/3} + O(1)$, $D_4(m) = (24m)^{1/4} + O(1)$.

Proof. Write $x = (k!\, m)^{1/k}$ and $D = D_k(m)$; by Lemma 1, D is the least integer d with $f(d,k) \geq m$.

Lower bound on D. Since $\binom{d}{i} \leq d^i/i!$,

$$f(d,k) \leq \sum_{i=1}^{k} \frac{d^i}{i!} \leq \frac{(d+k)^k}{k!}.$$

(Indeed, by the binomial theorem $\frac{(d+k)^k}{k!} = \sum_{i=0}^{k} \frac{d^i}{i!} \cdot \frac{k^{k-i}}{(k-i)!}$, and $\frac{k^j}{j!} \geq 1$ for $0 \leq j \leq k$, so each term with $1 \leq i \leq k$ is at least $d^i/i!$.) Then $m \leq f(D,k) \leq (D+k)^k/k!$ gives $D \geq x - k$.

Upper bound on D. By minimality $f(D-1,k) < m$. If $D - 1 \geq k - 1$, then

$$f(D-1,k) \geq \binom{D-1}{k} = \frac{(D-1)(D-2)\cdots(D-k)}{k!} \geq \frac{(D-k)^k}{k!},$$

so $(D-k)^k < k!\, m$, i.e. $D < x + k$. If $D - 1 < k - 1$ then $D < k \leq x + k$ trivially. □

3 The 1–3 strategy with optimal intervals

We now design the first egg's intervals. Suppose we want to be able to finish within d drops in total. If the first egg breaks on its j-th drop ($j = 1, 2, \ldots, d$), then j drops have been used, and 3 eggs and $d - j$ drops remain. By Lemma 1, 3 eggs and $d - j$ drops can handle at most $f(d-j,3)$ testable floors, i.e. a block of at most $f(d-j,3) + 1$ possibilities for n. So the best we can do is to make the j-th interval exactly that big:

$$x_j - x_{j-1} = 1 + f(d-j,3), \qquad j = 1, 2, \ldots, d, \quad x_0 = 0. \tag{1}$$

Notice that the intervals *shrink* as j grows: later in the process there are fewer drops left for the 3-egg search, so the block it must search has to be smaller.

Theorem 1 (Optimal 1–3)**.** *The 1–3 strategy with intervals (1) finds n within d drops for every $N \leq f(d,4)$. Consequently the best 1–3 strategy uses exactly*

$$D_4(N) = \min\{d : \ f(d,4) \geq N\} = (24N)^{1/4} + O(1)$$

drops, which is the best possible for 4 eggs by any strategy whatsoever.

Proof. It works. Drop the first egg from $x_1, x_2, \dots$ (never above floor N). If it first breaks at x_j, then $n \in \{x_{j-1}+1, \dots, x_j\}$, so there are $f(d-j,3)$ testable floors $x_{j-1}+1, \dots, x_j-1$; with 3 eggs and $d-j$ drops left, Lemma 1 says we can finish. The total is $j+(d-j)=d$ drops. If the egg survives all of $x_1, \dots, x_d$ and $x_d \geq N$, we have $n = N+1$. So it suffices to check that $x_d \geq N$ whenever $N \leq f(d,4)$, which follows from

$$x_d = \sum_{j=1}^{d} \big(1 + f(d-j,3)\big) = d + \sum_{i=0}^{d-1} f(i,3) = d + \sum_{r=1}^{3} \sum_{i=0}^{d-1} \binom{i}{r} = d + \sum_{r=1}^{3} \binom{d}{r+1} = f(d,4). \quad (2)$$

Here we used the hockey-stick identity $\sum_{i=0}^{d-1} \binom{i}{r} = \binom{d}{r+1}$ (proof: induction on d using Pascal's rule, $\binom{d}{r+1} = \binom{d-1}{r+1} + \binom{d-1}{r}$).

It is optimal. By Lemma 1, no 4-egg strategy of any kind can handle more than $f(d,4)$ floors in d drops, and by (2) this strategy handles exactly that many. Hence its worst-case cost is $D_4(N)$, and Lemma 2 gives $D_4(N) = (24N)^{1/4} + O(1)$. □

Another way to read this: the recursion $f(d,4) = 1 + f(d-1,3) + f(d-1,4)$ of Lemma 1 says "drop an egg; if it breaks you have a 3-egg problem with one fewer drop; if it survives you have the same 4-egg problem with one fewer drop." Unrolling the second branch over and over is exactly the 1–3 strategy. So the 1–3 approach with optimal intervals *is* the optimal 4-egg strategy.

A worked example. Take $N = 100$. Since $f(7,4) = 98 < 100 \leq 162 = f(8,4)$, we need $d = 8$ drops. The intervals (1) are $1 + f(7,3), 1 + f(6,3), \dots = 64, 42, 26, 15, 8, 4, 2, 1$, whose sum is $162 = f(8,4)$. We drop the first egg from floor 64. If it breaks, $n \in \{1, \dots, 64\}$, i.e. $63 = f(7,3)$ testable floors, which 3 eggs handle in 7 more drops. If it survives, our next drop is from $\min(64+42, 100) = 100$. If it breaks there, $n \in \{65, \dots, 100\}$, i.e. $35 \leq f(6,3) = 41$ testable floors, handled by 3 eggs in 6 more drops (total $2+6=8$). If it survives, $n = 101$. The worst case is 8 drops.

Size of the intervals. The first interval is $1 + f(d-1,3) \approx d^3/6$ with $d \approx (24N)^{1/4}$, i.e. about $\frac{24^{3/4}}{6} N^{3/4} \approx 1.81\, N^{3/4}$ floors; the j-th interval is about $(d-j)^3/6$.

4 The 2–2 strategy

By Lemma 2,

$$T_{22}(N,F) = D_2(M-1) + D_2(F-1) = \sqrt{2(M-1)} + \sqrt{2(F-1)} + O(1), \qquad M = \left\lceil \frac{N+1}{F} \right\rceil.$$

Since $|M - 1 - N/F| \leq 1$ and $|\sqrt{a} - \sqrt{b}| \leq \sqrt{|a-b|}$, replacing $M-1$ by N/F and $F-1$ by F changes the total by at most an absolute constant. So, uniformly in N and F,

$$T_{22}(N,F) = \sqrt{\frac{2N}{F}} + \sqrt{2F} + O(1). \quad (3)$$

The two terms compete: a large F makes the first stage cheap and the second stage expensive, and vice versa. Let $h(F) = \sqrt{2}\,(N^{1/2}F^{-1/2} + F^{1/2})$. Then

$$h'(F) = \sqrt{2}\Big(-\tfrac{1}{2}N^{1/2}F^{-3/2} + \tfrac{1}{2}F^{-1/2}\Big) = \frac{\sqrt{2}}{2} F^{-3/2}\big(F - N^{1/2}\big),$$

which is negative for $F < \sqrt{N}$ and positive for $F > \sqrt{N}$. So the unique minimum is at $F = \sqrt{N}$, where each term equals $\sqrt{2}\,N^{1/4}$:

$$\min_F T_{22}(N,F) = 2\sqrt{2}\,N^{1/4} + O(1) \approx 2.8284\,N^{1/4}. \tag{4}$$

(A quicker view of the lower bound: by AM–GM, $\sqrt{2N/F} + \sqrt{2F} \geq 2\sqrt{\sqrt{2N/F}\sqrt{2F}} = 2\sqrt{2\sqrt{N}} = 2\sqrt{2}\,N^{1/4}$, with equality iff $F = \sqrt{N}$. Rounding F to an integer costs only $O(1)$.)

5 The comparison

Theorem 2. *(a) For* every $N \geq 1$ *and every superfloor size* $F \geq 1$*,*

$$D_4(N) \;\leq\; T_{22}(N,F),$$

i.e. the 1–3 *approach (with optimal intervals) never needs more drops than the* 2–2 *approach.*

(b) For large N *it is strictly better:*

$$\min_F T_{22}(N,F) - D_4(N) = \left(2\sqrt{2} - 24^{1/4}\right)N^{1/4} + O(1) \approx 0.615\,N^{1/4},$$

and

$$\frac{\min_F T_{22}}{D_4(N)} \;\longrightarrow\; \left(\frac{64}{24}\right)^{1/4} = \left(\frac{8}{3}\right)^{1/4} \approx 1.278 \qquad (N \to \infty).$$

Proof. (a) The 2–2 approach is one particular strategy for 4 eggs, and by Theorem 1 the 1–3 approach with optimal intervals is optimal among all of them.

(b) By Theorem 1 and (4) the two costs are $24^{1/4}N^{1/4} + O(1)$ and $2\sqrt{2}\,N^{1/4} + O(1)$. Compare the constants via their fourth powers (both are positive):

$$\left(2\sqrt{2}\right)^4 = 64, \qquad \left(24^{1/4}\right)^4 = 24.$$

Since $64 > 24$, we get $2\sqrt{2} > 24^{1/4}$, and the ratio of constants is $(64/24)^{1/4} = (8/3)^{1/4}$. Since the difference of the leading terms grows like a positive constant times $N^{1/4}$ while the error terms are bounded, the inequality is strict for all sufficiently large N. □

So the two "fourth-power constants" are 64 for 2–2 and 24 for 1–3; smaller is better.

6 A second derivation: counting floors per drop

Here is a cleaner way to see where 64 and 24 come from, and why 1–3 wins. Suppose we are allowed d drops in total. Roughly (ignoring lower-order terms, by Lemma 1) a drops with k eggs can resolve $a^k/k!$ floors.

- 2–2**:** the two stages *multiply*: the number of superfloors times the size of a superfloor is the number of floors. Spending a drops on the superfloors and $b = d - a$ inside one superfloor, we resolve about $\frac{a^2}{2} \cdot \frac{b^2}{2} = \frac{a^2b^2}{4}$ floors. This is maximized at $a = b = d/2$ (by AM–GM or calculus), giving $\frac{d^4}{64}$.

- 1–3 **with optimal intervals:** the stages *add up* over the possible breaking times. The block examined when the first egg breaks on drop j has about $\frac{(d-j)^3}{6}$ floors, so the total is

$$\sum_{j=1}^{d} \frac{(d-j)^3}{6} \;\approx\; \int_0^d \frac{t^3}{6}\,dt = \frac{d^4}{24}.$$

  (This is the continuous version of the exact identity (2).)

Setting the number of floors equal to N and solving for d gives $d \approx (64N)^{1/4} = 2\sqrt{2}\,N^{1/4}$ for 2–2 and $d \approx (24N)^{1/4}$ for 1–3, in agreement with (4) and Theorem 1. Since $\frac{1}{24} > \frac{1}{64}$, d drops go further with 1–3.

Why does 1–3 win? In the 2–2 approach the drop budget d is split once and for all between the two stages (a drops for the superfloors, $b = d - a$ inside one), and all superfloors have the same size F, sized for b drops. But if the superfloor stage happens to finish after fewer than a drops, the leftover budget is simply wasted, because the superfloor was not allowed to be any bigger. The 1–3 strategy with optimal intervals avoids this: the block reached when the first egg breaks on drop j is sized for exactly the $d - j$ drops that remain (early blocks are huge, late blocks small), so no drop in the budget is wasted on any branch. This is precisely what the capacity recursion of Lemma 1 prescribes, which is why 1–3 cannot be beaten.

Remark (Equal intervals)**.** If one insisted on equal intervals G for the first egg, the cost would be $\frac{N}{G} + (6G)^{1/3} + O(1)$. Writing $G = cN^{3/4}$ and minimizing $\frac{1}{c} + 6^{1/3}c^{1/3}$ (or by AM–GM on the four terms $\frac{N}{G}$ and $3 \times \frac{1}{3}(6G)^{1/3}$) gives $4(2/9)^{1/4}N^{1/4} \approx 2.746\,N^{1/4}$, at $G \approx 1.4565\,N^{3/4}$. This is still better than 2–2 (since $512/9 < 64$) but worse than the optimal intervals (since $512/9 > 24$). The picture in constants (fourth powers; smaller is better) is therefore

$$\underbrace{24}_{\text{1–3, optimal intervals}} \;<\; \underbrace{512/9 \approx 56.9}_{\text{1–3, equal intervals}} \;<\; \underbrace{64}_{\text{2–2}}.$$

7 Numerical check

We computed exact costs from Lemma 1: $D_k(m)$ is the smallest d with $\sum_{i=1}^{k} \binom{d}{i} \geq m$, $T_{22}(N, F) = D_2(M-1) + D_2(F-1)$ with $M = \lceil (N+1)/F \rceil$, minimized over F (one minimizer is shown). We also ran the 1–3 strategy with intervals (1) explicitly for several N (every n from 1 to $N+1$) and confirmed that its worst case equals $D_4(N)$, and we confirmed Lemma 1 against a brute-force minimax search for $k \leq 4$ and $m < 80$.

N	best F	2–2 drops	1–3 drops (optimal)	1–3 (equal intervals)
10^2	7	8	8	8
10^3	22	15	13	15
10^4	65	28	23	27
10^6	816	89	71	86
10^8	8 488	283	222	274
10^{10}	96 394	894	701	868
10^{12}	984 436	2 828	2 214	2 746
prediction (drops/$N^{1/4}$)		2.8284	2.2134	2.7464

At $N = 10^{12}$ we have $N^{1/4} = 1000$, so the drop counts are essentially 1000 times the predicted constants, and the best F is about $\sqrt{N}$, as predicted. For every $N \le 6000$ we computed the best 2–2 cost and $D_4(N)$: the 1–3 approach was never worse (as Theorem 2(a) guarantees), strictly better for 5900 values of N, and tied for the other 100 (all with $N \le 175$). The ratio $2828/2214 \approx 1.277$ at $N = 10^{12}$ matches $(8/3)^{1/4} \approx 1.278$.

8 Conclusion

With 4 eggs and N floors, the best 2–2 approach takes about $2\sqrt{2}\,N^{1/4} \approx 2.83\,N^{1/4}$ drops (superfloors of size $F \approx \sqrt{N}$). The 1–3 approach, with the first egg dropped at the optimal (shrinking) intervals $1 + f(d-j, 3)$ and the best 3-egg strategy after it breaks, takes $24^{1/4}N^{1/4} \approx 2.21\,N^{1/4}$ drops, which is the optimum over *all* four-egg strategies. So 1–3 is never worse than 2–2, and for large N it is better by a factor of $(8/3)^{1/4} \approx 1.28$.

References

[1] S. J. Miller, *Egg Drop Mathematics: It IS all its cracked up to be!*, talk slides, Williams College, https://web.williams.edu/Mathematics/sjmiller/public_html/math/talks/eggdropmath_Hampshire28Jul2026.pdf; video: https://youtu.be/98cpdt2EsCM.

[2] *Egg drop problems*, paper in the Fibonacci Quarterly (see the link for authors), https://web.williams.edu/Mathematics/sjmiller/public_html/math/papers/EggDropProblems_FibonacciQuarterly.pdf.